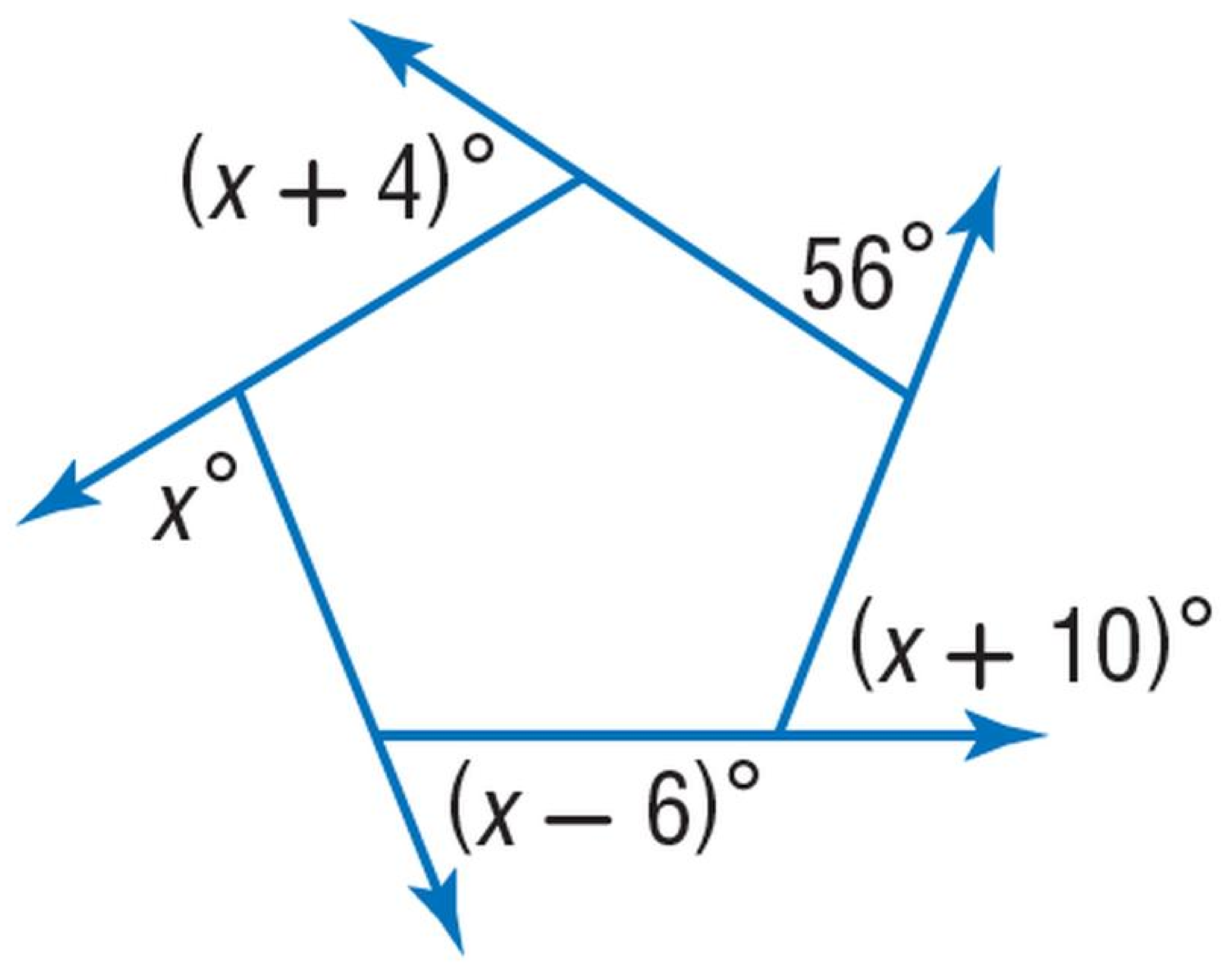
From Parsed Structure to Compact Theorem Guidance

A compact route works best when it is target-bound, checkable, and short enough to steer the solver without overpowering it.

Raw diagram + question



The image alone gives geometry cues, but not the preferred action.

Compact theorem route

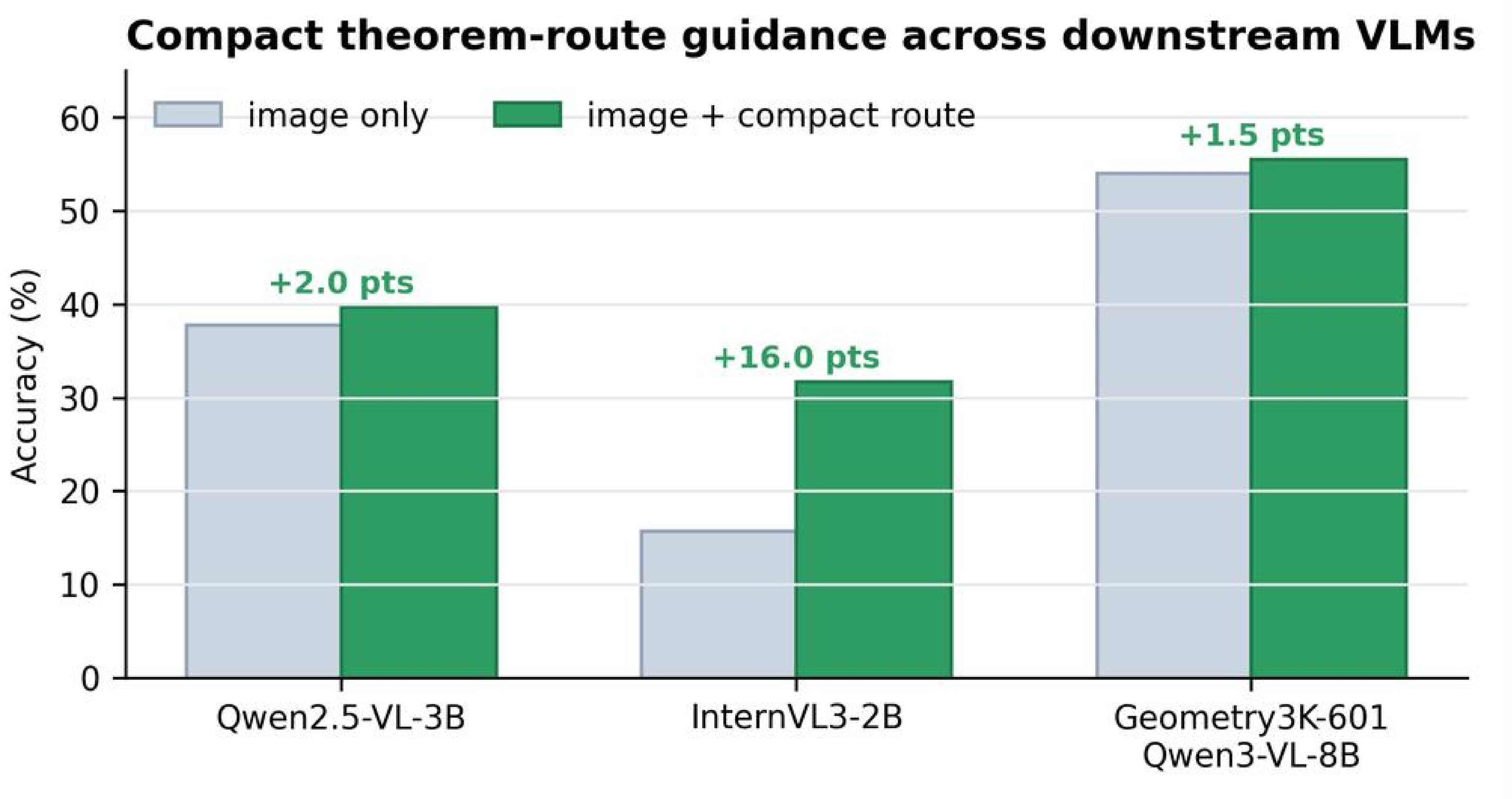
1. Use midpoint / midsegment relation
2. Bind the relation to the target side
3. Compute from the matched ratio

Short route = theorem-level action, not a full natural-language solution.

Why it helps

- Compress structure into theorem actions
- Bind the route to the target quantity
- Verify before trusting the hint

Downstream accuracy



Cross-model gain

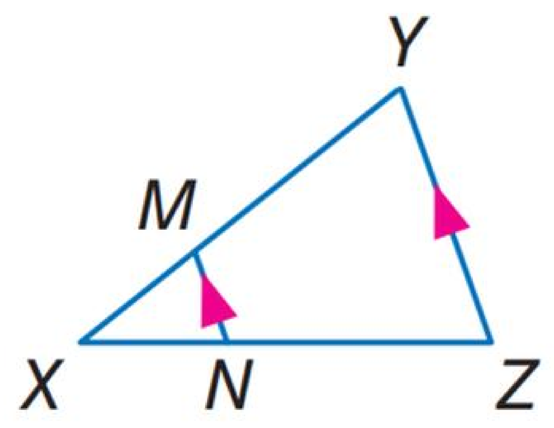
Qwen2.5-VL-3B: 37.75% -> 39.75%

InternVL3-2B: 15.75% -> 31.75%

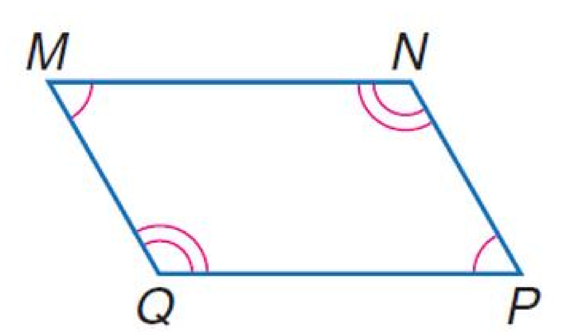
Qwen3-VL-8B: 54.08% -> 55.57%

Route can help or harm

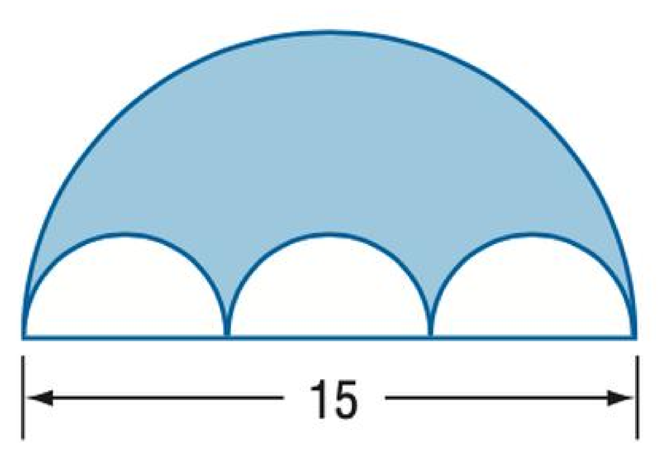
Win: route targets similarity
parallel angles -> similar ratio



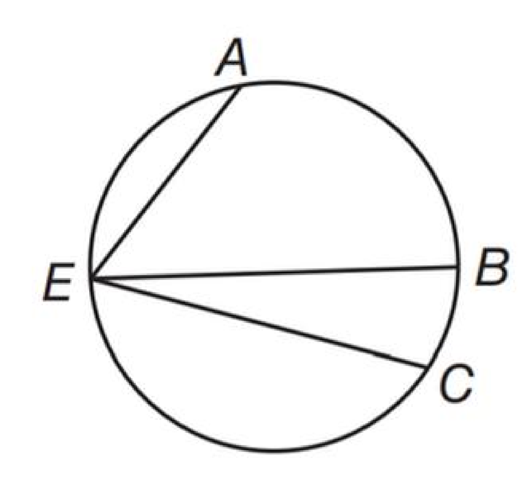
Loss: true theorem, wrong target
opposite-angle cue distracts



Win: coarse route activates formula
circle/semicircle area



Loss: over-specific route
arc theorem distracts



Key point: compact routes are useful when they expose target-relevant theorem actions; the next step is calibrated candidate exposure.

Pairwise diagnosis

Win = raw wrong -> exposed correct

Loss = raw correct -> exposed wrong

The route is useful only when it points to the target quantity.
If it points to the wrong theorem or wrong binding, it can flip a correct answer.

Selective exposure > blind trust